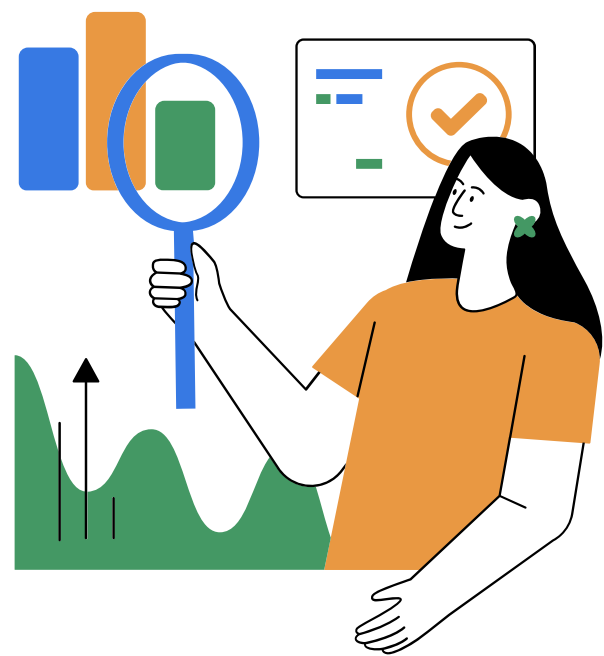


GEN-AI Powered

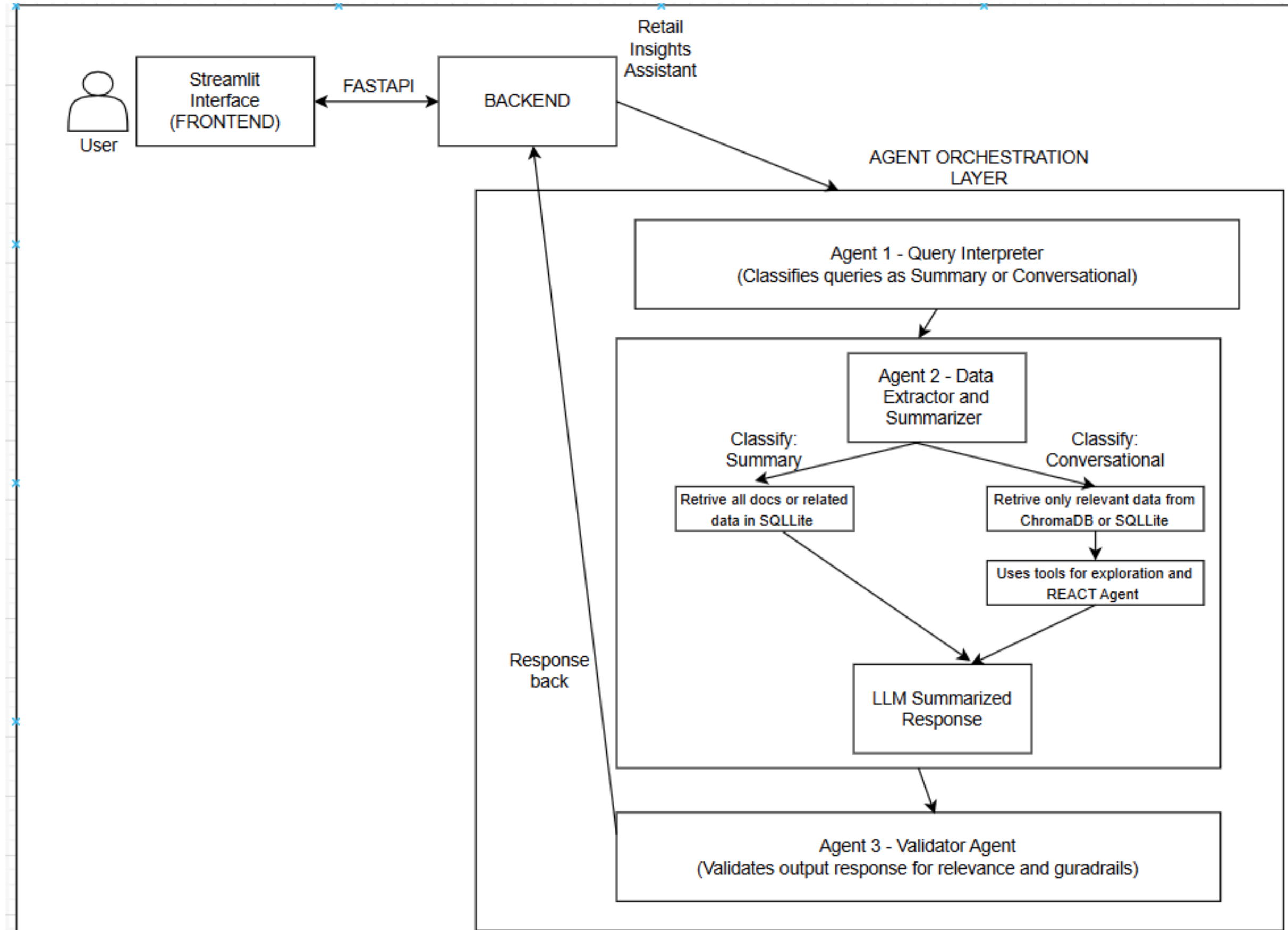
RETAIL INSIGHT ASSISTANT

Enterprise Architecture for Large-Scale AI-Powered Analytics

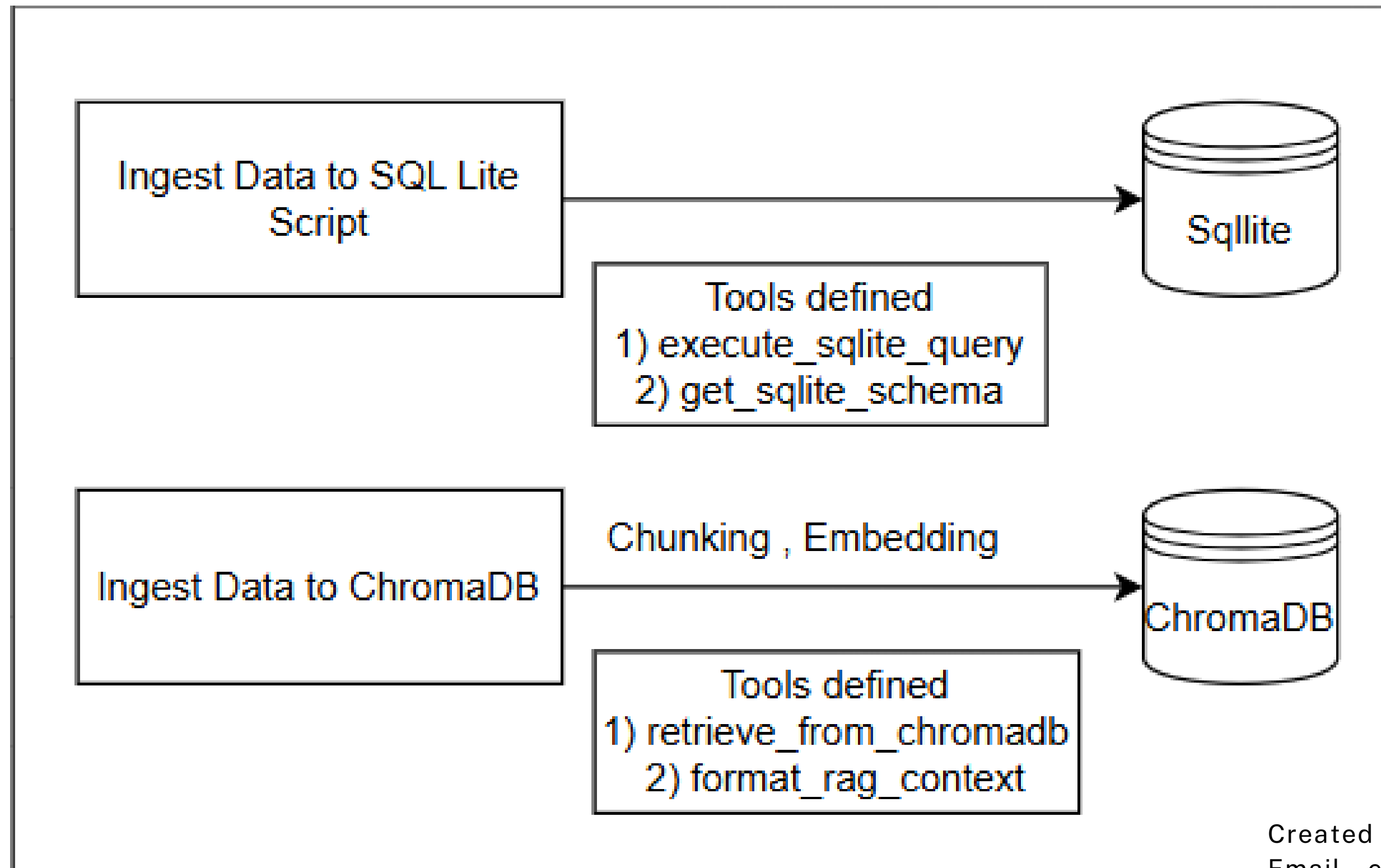


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CURRENT ARCHITECTURE



TOOLS CALLING AND STORAGE





RETAIL INSIGHTS ASSISTANT

<<

System Status

API: Offline

Data: No Data

Upload Data

Upload

200MB per file • CSV, XLSX, XLS

Powered by LangChain + Gemini

Deploy

Retail Insights Assistant

AI-Powered Sales Analytics

No data loaded. Upload a CSV file using the sidebar to get started.

Ask Your Data

e.g., Top 5 products by sales?

Ask

Generate Summary

Clear Chat

Try asking:

What are the top 5 products by sales?

Show revenue by region

Average order value?

System Status

API: Online

Data: Ready

Records: 128,975

Upload Data

Upload

200MB per file • CSV, XLSX, XLS

Powered by LangChain + Gemini

Deploy

Retail Insights Assistant

AI-Powered Sales Analytics

128,975

Records

24

Columns

3

AI Agents

Ask Your Data

You: What are the top 5 products by sales?

Assistant:

Direct Answer The top 5 products by sales are dominated by two main styles: the "J0230 Set" (sizes M and S) and the "JNE3797 Western Dress" (sizes L, M, and S), generating a combined total of "\$42,39M" in revenue.

Data Evidence

Rank	SKU	Category	Size	Total Quantity Sold	Total Sales (€)	Average Unit Price (€)
1	J0230-SM-M	Set	M	468	527,699.20	1,127.56
2	JNE3797-WB-L	Western Dress	L	661	524,581.77	793.62
3	J0230-SM-S	Set	S	421	479,937.14	1,139.99
4	JNE3797-WB-M	Western Dress	M	561	454,290.16	809.79
5	JNE3797-WB-S	Western Dress	S	503	407,302.57	809.75

Insight Layer

- High-Value vs. High-Volume Drivers:** The J0230 Set acts as a high-value driver with an average unit price of €1,139.99, 46% higher than the store's overall AUP of €809.75. Conversely, the JNE3797 Western Dress acts as a volume driver, selling 1,725 units across its top three sizes at a lower price point (€793.62).
- Size Demand Concentration:** Medium (M) and Large (L) sizes represent the highest revenue-generating segments for these top products, indicating strong product-market fit within standard sizing.
- Product Line Dominance:** Just two product designs (across different sizes) account for 3.05% of the total store revenue (€78.59M), highlighting a high concentration risk but also a clear customer preference.

KPI Snapshot

- Total Store Revenue: €78,592,678.30
- Total Orders: 120,378
- Average Order Value (AOV): €652.88
- Top 5 Products Revenue Contribution: €79,903,610.64 (1% of total revenue)



RETAIL INSIGHTS ASSISTANT

System Status

API: Online

Data: Ready

Records: 128,975

Upload Data

Upload

100MB per file • CSV, XLS, XLSX

Powered by

LangChain

• Gemini

50%

Reset

Retail Insights Assistant

AI-Powered Sales Analytics

128,975

Records

24

Columns

3

AI Agents

Ask Your Data

You: What are the top 5 products by sales?

Assistant:

Direct Answer The top 5 products by sales are dominated by two main styles: the **"J0230 Set"** (sizes M and S) and the **"JNE3797 Western Dress"** (sizes L, M, and S), generating a combined total of **"¥2.39M"** in revenue.

Data Evidence

Rank	SKU	Category	Size	Total Quantity Sold	Total Sales (¥)	Average Unit Price (¥)
1	J0230-SKD-M	Set	M	468	527,699.20	1,127.56
2	JNE3797-KR-L	Western Dress	L	661	524,581.77	793.62
3	J0230-SKD-S	Set	S	421	479,937.14	1,139.99
4	JNE3797-KR-M	Western Dress	M	561	454,290.16	809.79
5	JNE3797-KR-S	Western Dress	S	503	407,302.57	809.75

Insight Layer

- High-Value vs. High-Volume Drivers:** The **J0230 Set** acts as a high-value driver with an average unit price of **¥1,133** (+73.5% higher than the store's overall AOV of **¥652.88**). Conversely, the **JNE3797 Western Dress** acts as a volume driver, selling 1,725 units across its top three sizes at a lower price point (¥804).
- Size Demand Concentration:** Medium (M) and Large (L) sizes represent the highest revenue-generating segments for these top products, indicating strong product-market fit within standard sizing.
- Product Line Dominance:** Just two product designs (across different sizes) account for 3.05% of the total store revenue (¥78.59M), highlighting a high concentration risk but also a clear customer preference.

Ask Your Data

You: What are the top 5 products by sales?

Assistant:

Direct Answer The top 5 products by sales are dominated by two main styles: the **"J0230 Set"** (sizes M and S) and the **"JNE3797 Western Dress"** (sizes L, M, and S), generating a combined total of **"¥2.39M"** in revenue.

Data Evidence

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KPI Snapshot

- Total Store Revenue:** ¥78,592,678.30
- Total Orders:** 120,378
- Average Order Value (AOV):** ¥652.88
- Top 5 Products Revenue Contribution:** ¥2,393,810.84 (3.05% of total revenue)
- Total Units Sold (Top 5):** 2,614 units

Recommendation

Optimize Inventory & Bundling: Implement



Blend

Summary

RETAIL INSIGHTS ASSISTANT

You: Generate a comprehensive summary

🗨 Assistant:

Based on the provided dataset of **128,975 records** from the SQLite database, here is a comprehensive analytical summary.

Note: Since no unstructured documents were available in the ChromaDB database, this summary focuses on the deep-dive analysis of the structured transactional data.

1. Key Insights from the Structured Sales Data

The sample of 50 records reveals several crucial characteristics of the sales operations, product performance, and customer behavior:

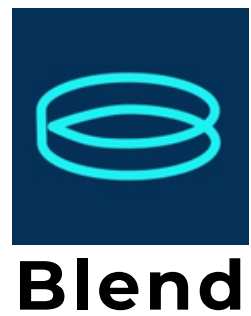
- Product Catalog & Demand Categories:
 - The inventory is primarily focused on Indian ethnic and fusion wear for women and men.
 - "Kurta" and "Set" (kurta-pant/dupatta sets) are the dominant product categories, followed by "Western Dress", "Top", and "Ethnic Dress".
 - There is a highly inclusive size range, stretching from XS to 6XL (including plus sizes like 3XL, 4XL, and 6XL).
- Revenue & Pricing Structure:
 - All transactions are conducted in Indian Rupees (INR).
 - Item prices in the sample range from ₹299.00 (for basic Kurtas) to ₹1,233.00 (for Sets), indicating an affordable, mid-range fashion positioning.
- Geographic Footprint:
 - The business has a truly pan-India reach, with a heavy concentration of deliveries in major Tier-1 and Tier-2 urban hubs, including Mumbai (Maharashtra), Bengaluru (Karnataka), Chennai (Tamil Nadu), Hyderabad (Telangana), and Lucknow (Uttar Pradesh).
- Fulfillment & Shipping Dynamics:
 - Fulfillment Channels: The business uses a hybrid model:
 - Amazon Fulfilled (FBA): Generally associated with an "Expedited" shipping service level.
 - Merchant Fulfilled (MFN): Managed via "Easy Ship" with a "Standard" shipping service level.
 - Promotional Drivers: Promotions heavily drive sales. Two main promotion types recur:
 - IN Core Free Shipping (highly associated with Amazon-fulfilled orders).
 - Amazon PLCC Free-Financing Universal Merchant (associated with Merchant-fulfilled orders).
- Order StatUSES & Cancellations:
 - A noticeable number of orders in the sample have a status of "Cancelled".
 - Interestingly, for some cancelled orders (e.g., Index 0, Index 3, Index 23), the quantity is 8 but an Amount is still recorded. For others (like Index 8 and Index 29), both Qty and Amount are null/zero. This suggests different stages of cancellation (pre-shipment vs. transit return).
- B2B vs. B2C Market Share:
 - The vast majority of transactions are direct-to-consumer (B2C). However, there is active, minor penetration into B2B e-commerce (e.g., Index 2, an expedited order for a Kurta to Navi Mumbai).

2. Insights from Unstructured Documents

- No unstructured documents were available in ChromaDB for this analysis. Consequently, internal company memos, customer reviews, or external market reports were not factored in.

3. Overall Trends and Patterns

- The FBA Customer Experience Advantage: Amazon-fulfilled orders almost exclusively leverage "Expedited" shipping. This likely correlates with higher conversion rates and lower drop-offs compared to the Standard Merchant-fulfilled (Easy Ship) orders.



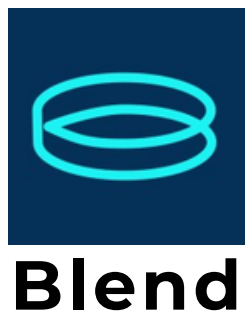
ARCHITECTURE SCALE

The current architecture and solution created working for limited number of files (csv, text, json) and for small number of records.

But similar architecture can be scaled to handle large volume of data like 100+ GBs of data at scale with few changes considering the scalability factor

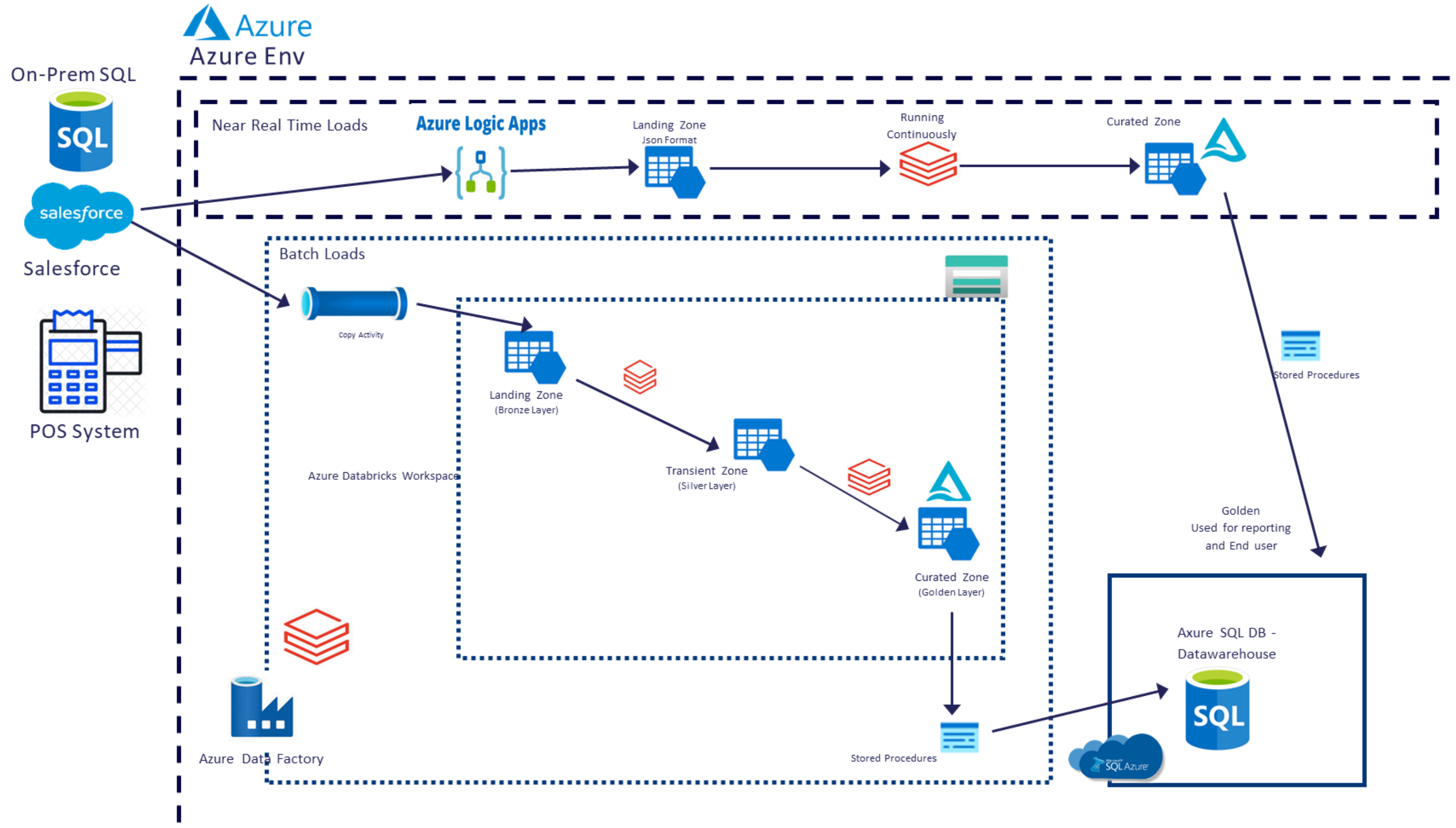


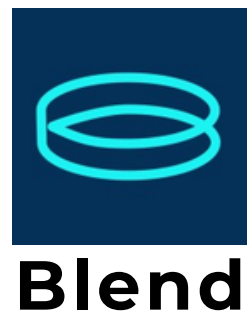
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DATA INGESTION - SQL

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DATA INGESTION - SQL

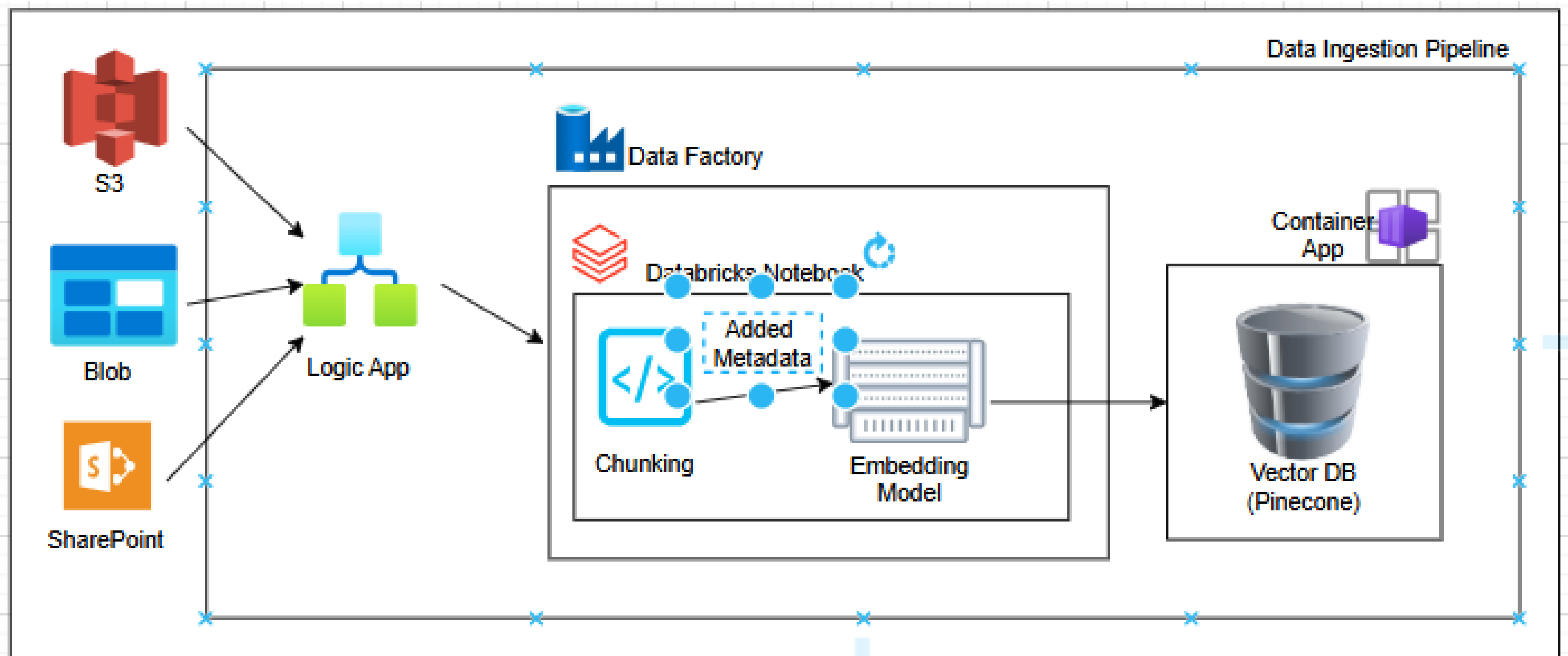
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Optimizing Data Storage in Azure SQL Server

- 1) Create domain specific data mart eg. Sales, Procurement and access level defined on user and data mart level
- 2) Clustered Columnstore Index (CCI) on large fact tables
- 3) Non-clustered indexes on RAG filter columns (doc_id, category, date)
- 4) Partition tables by date or category, pruning irrelevant data at query time
- 5) Materialized views for frequently accessed RAG contexts

DATA INGESTION - VECTORDB

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Blend

DATA INGESTION - VECTORDB

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Optimizing Pinecone Vector DB

1) Indexing

- Use HNSW index, best recall/speed tradeoff for RAG
- Set ef_construction high during build, lower ef at query time for speed

2) Metadata Filtering (Critical)

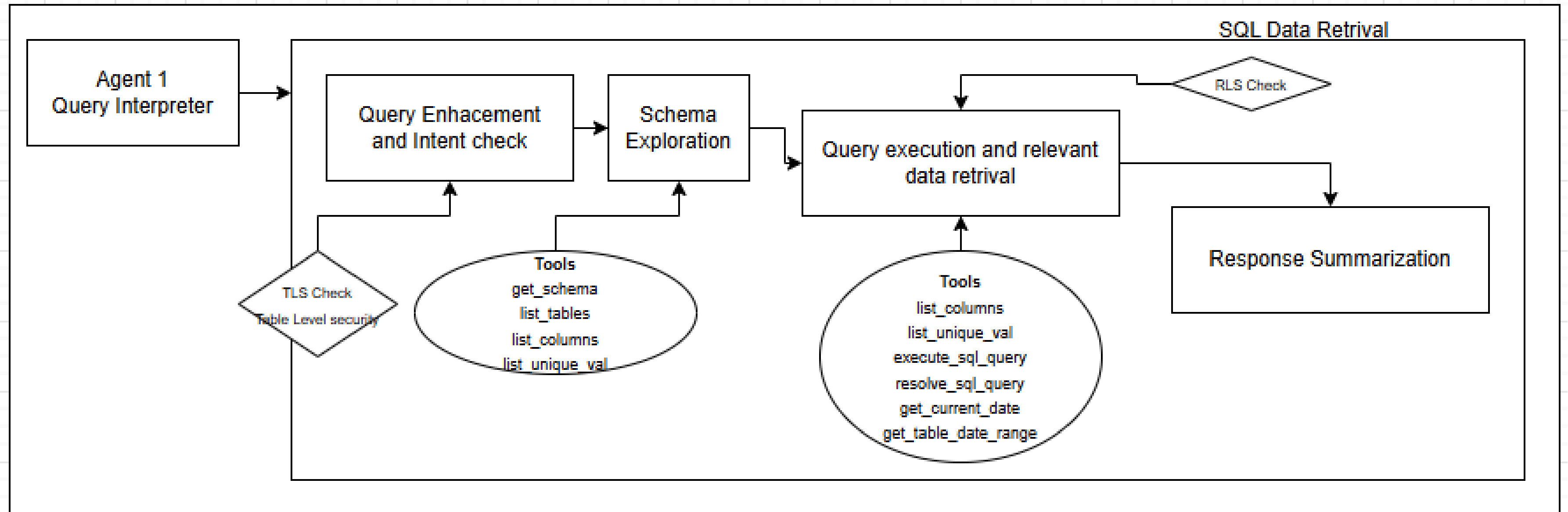
- Store rich metadata per chunk: source, doc_type, date, domain, chunk_id
- Always pre-filter by metadata before vector search, reducing search space massively

3) Fixed 512 tokens + 50 token overlap, optimal for most embedding models

4) Use Pinecone Serverless — auto-scales, no pod management

5) Add upsert deduplication logic in Databricks notebook before push.

RAG ARCHITECTURE - SQL

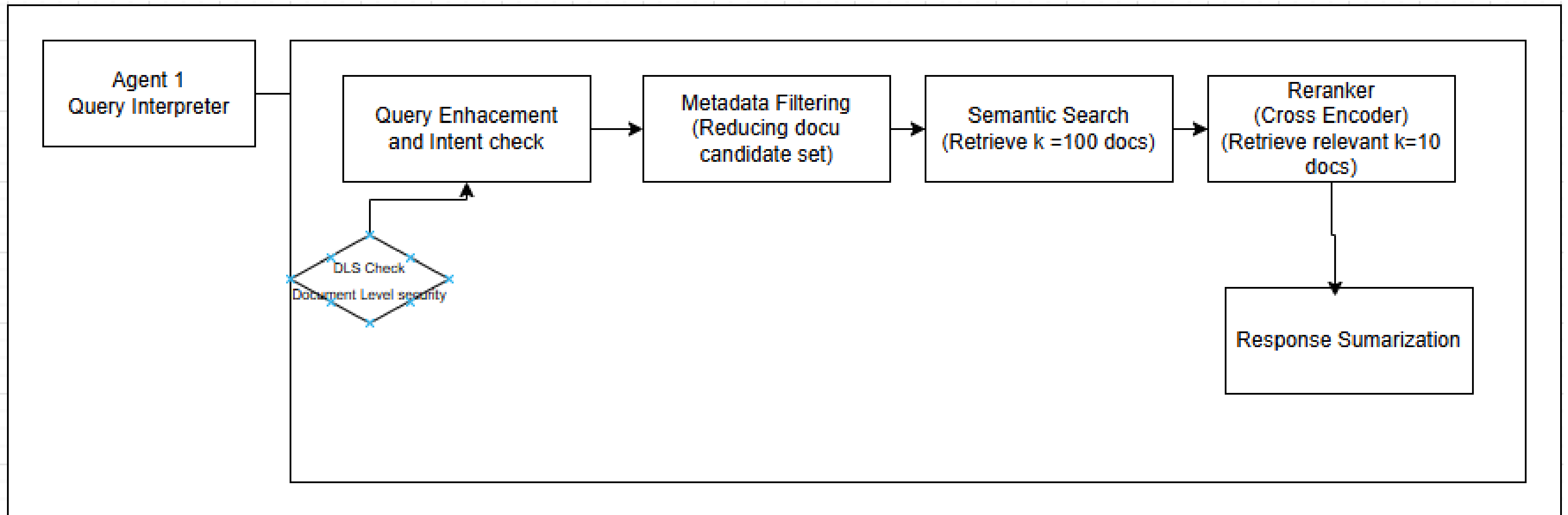




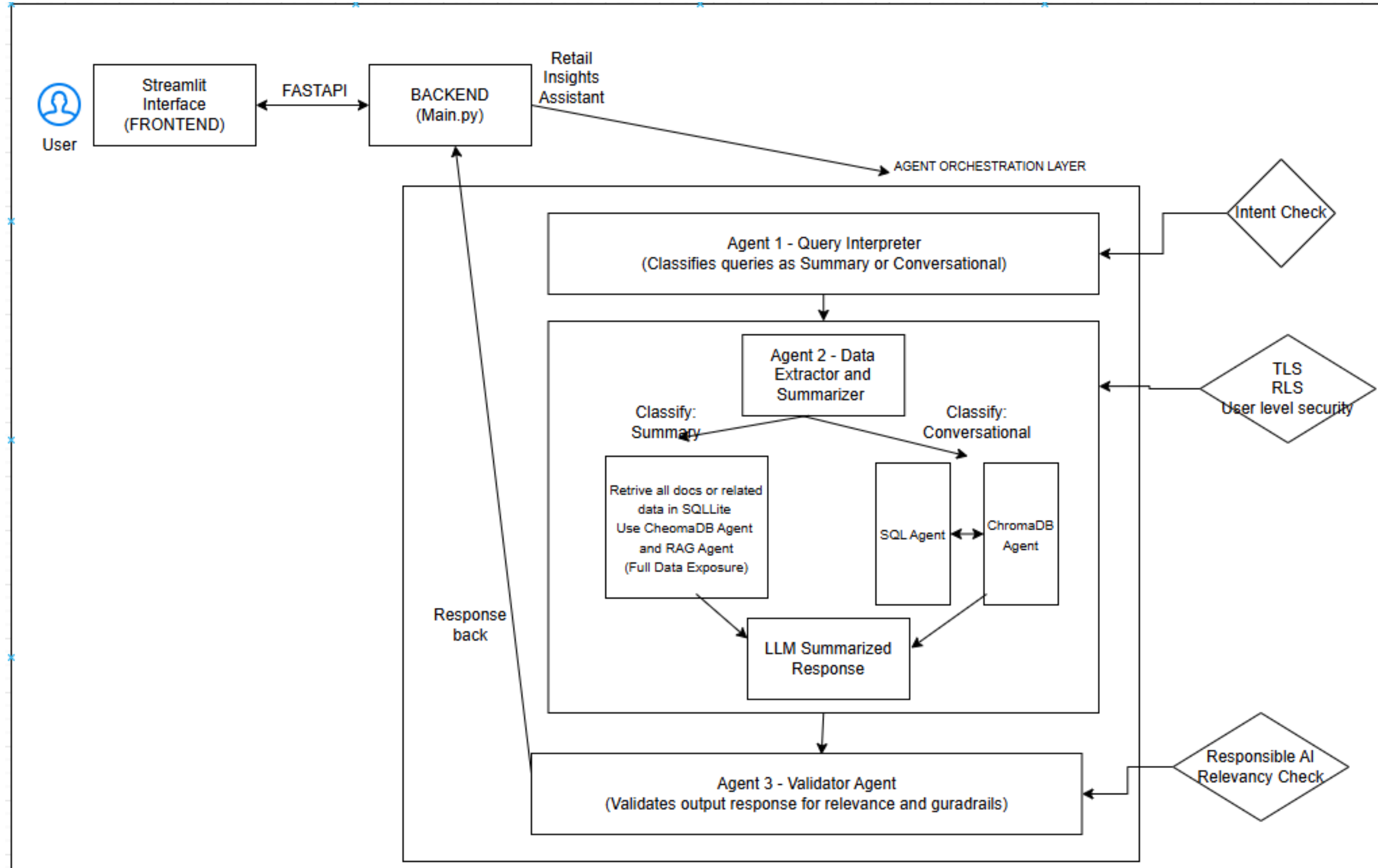
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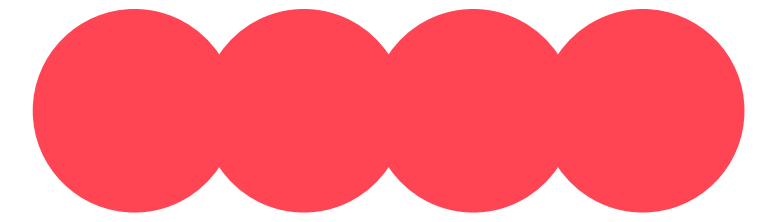
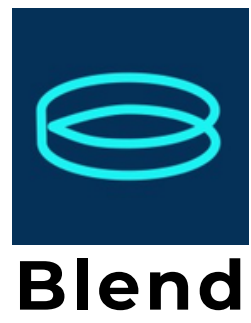
RAG ARCHITECTURE - VECTOR DB

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QUERY - RESPONSE PIPELINE





Thank You



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